ENGINEERING & DATA SCIENCES · CHEMISTRY S1-1

Exam practice

The atom and its quantum model

Handout chapters 1–3 · worked answers

25 questions · Moderate and Hard ★ only

How to use it

- Every question is followed by a worked answer in a blue box. Check the method (formula in letters first, then numbers with units) as well as the result.
- Moderate: one or two steps with a formula from the handout. Hard: several steps, reasoning in reverse, or ideas from two chapters at once.

Data $c = 3.00 \times 10^{8}\,\mathrm{m/s}$ $h = 6.62 \times 10^{-34}\,\mathrm{J \cdot s}$ $e = 1.6 \times 10^{-19}\,\mathrm{C}$ $1\ \mathrm{eV} = 1.6 \times 10^{-19}\,\mathrm{J}$ $N_A = 6.022 \times 10^{23}/\mathrm{mol}$ $R_H = 3.29 \times 10^{15}\,\mathrm{Hz}$ $a_0 = 52.9\ \mathrm{pm}$ $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^{9}\,\mathrm{N \cdot m^2/C^2}$ $m_e = 9.1094 \times 10^{-31}\,\mathrm{kg}$ $m_p = 1.0073\ \mathrm{amu}$ $m_n = 1.0087\ \mathrm{amu}$ $1\ \mathrm{amu} = 1.6605 \times 10^{-27}\,\mathrm{kg}$ $g = 9.81\,\mathrm{m/s^2}$ $E_n = -13.6\,Z^2/n^2\ \mathrm{eV}$.

Map of the question types

Chapter	Type of question	Questions
1	Counting protons, neutrons, electrons; isotopes and isoelectronic species	A1
1	Mass defect and energy of a nucleus	A2
1	Millikan: finding e from data; balancing a charged drop	A3
1	Rutherford: closest approach, density and size of the nucleus	A4
1	Thomson: comparing charge-to-mass ratios	A5
2	Converting between λ, ν, wavenumber, eV and $\mathrm{kJ\,mol^{-1}}$	B1, B7
2	Identifying a transition from a wavelength	B2
2	Which photons are absorbed; ionisation and the ejected electron	B3, D3
2	Emission cascade: number of lines, extreme wavelengths, visible lines	B4
2	Data analysis: Rydberg's constant from measured lines	B5
2	Hydrogen-like ions; coinciding lines	B6, D2
2	Explaining spectra (lines vs bands, emission vs absorption, colour)	B8
3	Allowed sets of quantum numbers; naming orbitals	C1
3	Counting orbitals and electrons with conditions	C2
3	Nodes; recognising an orbital from its nodes and planes	C3, C8
3	Radial probability density: maximum, probability inside a sphere	C4, C5
3	Degeneracy in H vs many-electron atoms; Madelung order	C6
3	Spin (sodium doublet) and Zeeman splitting	C7
3	Sizes $r = n^2 a_0/Z$	C9, D2
1–3	Synthesis: Coulomb energy and Bohr levels; unknown ion; photoionisation	D1–D3

Part A · The atom and its particles (chapter 1)

A1. Moderate Particle bookkeeping

(a) Complete the table.

Species	$^{37}_{17}\mathrm{Cl}^-$	$^{56}_{26}\mathrm{Fe}^{3+}$	$^{18}_{8}\mathrm{O}^{2-}$	$^{16}_{8}\mathrm{O}$	$^{23}_{11}\mathrm{Na}^+$
protons / neutrons / electrons					

(b) Which species are isotopes of each other? Which are isoelectronic, and with which noble gas? (c) A species has 26 protons, 30 neutrons and 23 electrons. Write its symbol in full.

Answer

(a) $^{37}_{17}\mathrm{Cl}^-$: 17 / 20 / 18. $^{56}_{26}\mathrm{Fe}^{3+}$: 26 / 30 / 23. $^{18}_{8}\mathrm{O}^{2-}$: 8 / 10 / 10. $^{16}_{8}\mathrm{O}$: 8 / 8 / 8. $^{23}_{11}\mathrm{Na}^+$: 11 / 12 / 10. (Neutrons $N = A - Z$; electrons $= Z -$ charge.)
(b) Isotopes: $^{18}\mathrm{O}$ and $^{16}\mathrm{O}$ (same $Z = 8$, different A); the charge does not matter. Isoelectronic: $^{18}_{8}\mathrm{O}^{2-}$ and $^{23}_{11}\mathrm{Na}^+$, 10 electrons each, like neon.
(c) $Z = 26$ is iron, $A = 26 + 30 = 56$, charge $26 - 23 = +3$: $^{56}_{26}\mathrm{Fe}^{3+}$.

A2. Hard ★ Where does the mass go? (helium-4)

A ^{4}He atom has a mass of 4.0026 amu. (a) Compute the total mass of its separate particles (2 protons, 2 neutrons, 2 electrons; $m_e = 1/1823$ amu). (b) Deduce the mass defect Δm in amu and in kg. (c) The binding energy is $E = \Delta m\, c^2$: compute it per atom, in J and MeV (1 MeV $= 1.6 \times 10^{-13}$ J), and per mole. Compare with a chemical bond (a few hundred kJ mol^{-1}).

Answer

(a) $2 \times 1.0073 + 2 \times 1.0087 + 2/1823 = 4.0331$ amu.
(b) $\Delta m = 4.0331 - 4.0026 = 0.0305$ amu $= 0.0305 \times 1.6605 \times 10^{-27} = 5.06 \times 10^{-29}$ kg.
(c) $E = 5.06 \times 10^{-29} \times (3.00 \times 10^8)^2 = 4.56 \times 10^{-12}$ J $= 28.5$ MeV per atom; per mole $4.56 \times 10^{-12} \times 6.022 \times 10^{23} = 2.74 \times 10^{12}$ J mol^{-1}, about ten million times a chemical bond energy: nuclear energies dwarf chemical ones.

A3. Hard ★ Millikan's experiment

(a) Charges measured on oil drops: 6.41, 9.62, 3.20, 12.82 and 4.81×10^{-19} C. Show that they are whole multiples of one charge and find it.
(b) A drop of oil (density 886 kg m^{-3}, radius 1.00 µm) is held still between two horizontal plates $d = 5.0$ mm apart when $U = 380$ V is applied: the electric force qU/d balances the weight mg. Find its charge and the number of extra electrons it carries.

Answer

(a) Dividing by the smallest value: $6.41/3.20 = 2.00$, $9.62/3.20 = 3.01$, $12.82/3.20 = 4.01$, $4.81/3.20 = 1.50$. The ratio 1.50 shows that 3.20 is itself $2e$: dividing everything by 1.60×10^{-19} gives 4, 6, 2, 8 and 3. All are whole multiples of $e = 1.60 \times 10^{-19}$ C.
(b) $m = \rho \frac{4}{3}\pi r^3 = 886 \times \frac{4}{3}\pi (1.00 \times 10^{-6})^3 = 3.71 \times 10^{-15}$ kg; $mg = 3.64 \times 10^{-14}$ N. $q = \dfrac{mg\,d}{U} = \dfrac{3.64 \times 10^{-14} \times 5.0 \times 10^{-3}}{380} = 4.79 \times 10^{-19}$ C $\approx 3e$: three extra electrons.

Unofficial study material

A4. Hard ★ Rutherford on copper

An α particle (charge $+2e$) with kinetic energy 7.7 MeV heads straight for a copper nucleus ($Z = 29$). (a) At its closest approach all its kinetic energy has become potential energy $k\,q_1 q_2/d$. Find d. (b) Why is d only an upper limit for the radius of the nucleus? (c) The copper nucleus ($A = 63$) has a radius of about 4.8 fm. Compute its density and compare the radius of the atom (128 pm) with that of its nucleus. (d) Use this ratio to explain why most α particles cross a metal foil undeflected.

Answer

(a) $E_k = 7.7 \times 1.6 \times 10^{-13} = 1.23 \times 10^{-12}$ J; $d = \dfrac{k\,(2e)(29e)}{E_k} = \dfrac{8.99 \times 10^{9} \times 58 \times (1.6 \times 10^{-19})^2}{1.23 \times 10^{-12}} = 1.08 \times 10^{-14}$ m $= 10.8$ fm.
(b) The particle turns back before touching the nucleus; a faster particle would get closer. The nucleus is at most this large.
(c) $m \approx 63 \times 1.6605 \times 10^{-27} = 1.05 \times 10^{-25}$ kg, $V = \frac{4}{3}\pi(4.8 \times 10^{-15})^3 = 4.6 \times 10^{-43}$ m^3, $\rho \approx 2.3 \times 10^{17}$ kg m^{-3}. $128 \times 10^{-12}/4.8 \times 10^{-15} \approx 2.7 \times 10^{4}$.
(d) The nucleus occupies a fraction $(1/2.7 \times 10^{4})^2 \sim 10^{-9}$ of the cross-section of the atom: almost every α particle meets only the light, spread-out electrons and goes straight.

A5. Moderate Thomson's specific charge

In a deflection tube, the deviation is proportional to q/m. Compute q/m for an electron, a proton and an α particle ($m_\alpha = 4.0015$ amu) and rank their deviations. By what factor is the electron deviated more than the α particle?

Answer

Electron: $1.6\times10^{-19}/9.1094\times10^{-31} = 1.76\times10^{11}$ C kg^{-1}. Proton: $1.6\times10^{-19}/(1.0073\times1.6605\times10^{-27}) = 9.57 \times 10^{7}$ C kg^{-1}. α: $2 \times 1.6 \times 10^{-19}/(4.0015 \times 1.6605 \times 10^{-27}) = 4.82 \times 10^{7}$ C kg^{-1}. Deviation: electron $\gg$ proton $>$ α; the electron is deviated $1.76 \times 10^{11}/4.82 \times 10^{7} \approx 3.6 \times 10^{3}$ times more (and in the opposite direction).

Part B · Light and the hydrogen spectrum (chapter 2)

B1. Moderate One photon, five units

A photon carries 2.48 eV. Give its wavelength, frequency, wavenumber (cm^{-1}), the energy of one mole of such photons (kJ mol^{-1}), and the region of the spectrum.

Answer

$E = 2.48 \times 1.6 \times 10^{-19} = 3.97 \times 10^{-19}$ J. $\lambda = hc/E = 5.01 \times 10^{-7}$ m $= 501$ nm (visible, blue-green). $\nu = E/h = 5.99 \times 10^{14}$ Hz. $1/\lambda = 2.00 \times 10^{6}$ m$^{-1} = 2.00 \times 10^{4}$ cm^{-1}. Per mole: $3.97 \times 10^{-19} \times 6.022 \times 10^{23} = 2.39 \times 10^{5}$ J mol$^{-1} = 239$ kJ mol^{-1}, the order of a chemical bond.

B2. Moderate Which transition?

Hydrogen emits a line at $\lambda = 1282$ nm. Using Rydberg's formula, show that it cannot belong to the Lyman or the Balmer series, and identify the transition.

Answer

$\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2} = \dfrac{\nu}{R_H} = \dfrac{c}{\lambda R_H} = \dfrac{3.00 \times 10^{8}}{1282 \times 10^{-9} \times 3.29 \times 10^{15}} = 0.0711.$
Lyman ($n_1 = 1$): $1/n_2^2 = 0.929$, $n_2 = 1.04$, not an integer. Balmer ($n_1 = 2$): $1/n_2^2 = 0.179$, $n_2 = 2.36$, no. Paschen ($n_1 = 3$): $1/n_2^2 = 0.111 - 0.0711 = 0.0400$, $n_2 = 5.00$. The line is $5 \to 3$ (infrared, as expected for $\lambda > 821$ nm).

B3. Moderate Photons hitting ground-state hydrogen

Hydrogen atoms in their ground state receive photons of (a) 12.75 eV, (b) 12.50 eV, (c) 13.22 eV, (d) 20.0 eV. For each, say whether it is absorbed and what happens. For (d), compute the speed of the electron that leaves.

Answer

Absorption needs $h\nu = E_n - E_1 = 13.6(1 - 1/n^2)$ exactly, or $h\nu \geq 13.6$ eV (ionisation). (a) $1 - 1/n^2 = 0.9375$, $n = 4$: absorbed, $1 \to 4$. (b) $n = 3.52$: not an integer, not absorbed. (c) $n = 5.98 \approx 6$: absorbed, $1 \to 6$ ($13.6 \times 35/36 = 13.22$ eV). (d) $20.0 > 13.6$: ionised, $E_k = 20.0 - 13.6 = 6.4$ eV $= 1.02 \times 10^{-18}$ J; $v = \sqrt{2E_k/m_e} = \sqrt{2 \times 1.02 \times 10^{-18}/9.1094 \times 10^{-31}} = 1.50 \times 10^6$ m s^{-1}.

B4. Hard ★ Emission cascade from $n = 5$

A gas of hydrogen atoms is excited to $n = 5$. (a) How many different lines can it emit? (b) How many of them are visible, and which? (c) Compute the shortest and the longest wavelengths emitted. (d) Draw the level diagram with the transitions, grouped into series.

Answer

(a) Any pair of levels among 1–5: $5 \times 4/2 = 10$ lines.
(b) Visible lines are Balmer lines ending on $n = 2$: $3 \to 2$ (657 nm), $4 \to 2$ (486 nm), $5 \to 2$ (434 nm): three lines.
(c) Shortest: largest energy gap, $5 \to 1$: $\lambda = c/[R_H(1 - 1/25)] = 95.0$ nm (UV). Longest: smallest gap, $5 \to 4$: $\lambda = c/[R_H(1/16 - 1/25)] = 4.05$ µm (IR).
(d) Lyman (to $n = 1$): 4 lines; Balmer (to 2): 3; Paschen (to 3): 2; Brackett (to 4): 1. Total 10.

B5. Hard ★ Measuring Rydberg's constant

The four visible lines of hydrogen are measured at 656.3, 486.1, 434.0 and 410.2 nm. (a) Show that Balmer's formula predicts a straight line when $1/\lambda$ is plotted against $1/n^2$, and give its slope and intercept. (b) Using a linear fit (or the first and last points), find R_H in m^{-1} and in Hz. (c) Deduce the ionisation energy of hydrogen in eV and kJ mol^{-1}, and the wavelength of the Balmer limit.

Answer

(a) $\dfrac{1}{\lambda} = \dfrac{R_H}{c}\left(\dfrac{1}{4} - \dfrac{1}{n^2}\right) = \dfrac{R_H}{4c} - \dfrac{R_H}{c}\dfrac{1}{n^2}$: a line of slope $-R_H/c$ and intercept $R_H/(4c)$, with $n = 3, 4, 5, 6$.
(b) Points $(1/n^2, 1/\lambda)$: $(0.1111, 1.5237)$, $(0.0625, 2.0572)$, $(0.0400, 2.3041)$, $(0.0278, 2.4378)$, with $1/\lambda$ in 10^6 m^{-1}. Least squares: slope $= -1.0972 \times 10^7$ m^{-1}, intercept 2.743×10^6 m^{-1} ($\times 4 = 1.097 \times 10^7$, consistent). So $R_H/c = 1.097 \times 10^7$ m^{-1} and $R_H = 3.29 \times 10^{15}$ Hz.
(c) $E_I = hR_H = hc \times 1.097 \times 10^7 = 2.18 \times 10^{-18}$ J $= 13.6$ eV, i.e. $2.18 \times 10^{-18} \times 6.022 \times 10^{23} = 1.31 \times 10^6$ J mol^{-1} $= 1312$ kJ mol^{-1}. Balmer limit: $\lambda = 4/(1.097 \times 10^7) = 365$ nm.

B6. Hard ★ The Li^{2+} ion

(a) Compute the ionisation energy of Li^{2+} in its ground state and the wavelength of its $2 \to 1$ line. In which region is it? (b) Show that the Li^{2+} transition $9 \to 6$ has exactly the wavelength of hydrogen's $3 \to 2$ line. (c) Generalise: which Li^{2+} transitions reproduce all the Balmer lines?

Answer

(a) $E_I = 13.6 \times 3^2 = 122.4$ eV. $\nu = 9R_H(1 - 1/4) = 2.22 \times 10^{16}$ Hz, $\lambda = 13.5$ nm: far UV (almost X-rays).
(b) Li^{2+}: $9\left(\frac{1}{36} - \frac{1}{81}\right) = \frac{9}{36} - \frac{9}{81} = \frac{1}{4} - \frac{1}{9}$, which is hydrogen's $\frac{1}{2^2} - \frac{1}{3^2}$: same frequency, same wavelength (657 nm).
(c) $9\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = \frac{1}{(n_1/3)^2} - \frac{1}{(n_2/3)^2}$: it equals $\frac{1}{4} - \frac{1}{m^2}$ for $n_1 = 6$ and $n_2 = 3m$. The transitions

$9 \to 6$, $12 \to 6$, $15 \to 6 \ldots$ reproduce the Balmer lines $m = 3, 4, 5 \ldots$

B7. Moderate Counting photons

A 60 W sodium lamp turns 30 % of its power into light at 589 nm. How many photons does it emit per second, and how many moles of photons per hour?

Answer

Light power $P = 0.30 \times 60 = 18$ W. One photon: $E = hc/\lambda = 3.37 \times 10^{-19}$ J. $N/t = 18/3.37 \times 10^{-19} = 5.34 \times 10^{19}$ photons s^{-1}; per hour $5.34 \times 10^{19} \times 3600/6.022 \times 10^{23} = 0.319$ mol of photons.

B8. Moderate Explaining spectra

(a) Why do the absorption and emission spectra of hydrogen show lines at exactly the same wavelengths? (b) Why does a solution of $KMnO_4$ show a broad absorption band around 530 nm rather than thin lines, and why does it look purple? (c) What experimental fact led Bohr to say that the energy of the atom is quantised?

Answer

(a) Both involve the same pairs of levels: a photon of energy $E_{n_2} - E_{n_1}$ is emitted when the electron drops and absorbed when it rises.
(b) In a molecule in solution, many closely spaced levels (vibrations, interactions with the solvent) merge into a continuous band. The solution absorbs green light (about 530 nm), so we see the complementary colour, purple/magenta.
(c) Hydrogen absorbs and emits only a few discrete wavelengths: the atom can only exchange definite amounts of energy, so its energy takes only definite values.

Part C · Quantum numbers and atomic orbitals (chapter 3)

C1. Moderate Allowed or not?

For each set (n, ℓ, m_ℓ, m_s), say whether it is allowed. If so, name the orbital; if not, say which rule is broken: (a) $(2, 2, 0, +\frac{1}{2})$; (b) $(3, 2, -2, -\frac{1}{2})$; (c) $(4, 0, 1, +\frac{1}{2})$; (d) $(5, 3, -3, +\frac{1}{2})$; (e) $(1, 0, 0, 0)$; (f) $(3, 1, -1, +\frac{1}{2})$.

Answer

(a) No: $\ell \le n - 1 = 1$. (b) Yes: 3d. (c) No: $|m_\ell| \le \ell = 0$. (d) Yes: 5f. (e) No: $m_s = \pm\frac{1}{2}$ only. (f) Yes: 3p.

C2. Moderate Counting with conditions

(a) For $n = 4$: list the subshells and give the numbers of orbitals and electrons. (b) How many electrons of an atom can have $n = 3$ and $m_\ell = +1$? (c) How many can have $n = 4$ and $m_s = +\frac{1}{2}$? (d) How many orbitals with $\ell = 2$ exist for $n \le 5$?

Answer

(a) 4s, 4p, 4d, 4f: $1 + 3 + 5 + 7 = 16 = n^2$ orbitals, $2n^2 = 32$ electrons.
(b) $m_\ell = +1$ exists for 3p and 3d: 2 orbitals, 4 electrons.
(c) 16 orbitals, one electron with $m_s = +\frac{1}{2}$ each: 16.
(d) $\ell = 2$ needs $n \ge 3$: 3d, 4d, 5d, $3 \times 5 = 15$ orbitals.

C3. Moderate Nodes

(a) Give the numbers of radial, angular and total nodes of 2s, 3p, 4d, 5f and 4s. (b) Which orbital has 2 radial nodes and 1 angular node? Which has no radial node and 3 angular nodes? (c) Can an orbital of the shell $n = 4$ have 2 radial and 2 angular nodes?

Answer

(a) Radial $n - \ell - 1$, angular ℓ, total $n - 1$: 2s 1/0/1; 3p 1/1/2; 4d 1/2/3; 5f 1/3/4; 4s 3/0/3.
(b) $\ell = 1$, $n = 2 + 1 + 1 = 4$: 4p. $\ell = 3$, $n = 4$: 4f.
(c) No: the total would be 4, but a shell $n = 4$ has $n - 1 = 3$ nodes. It is a 5d orbital.

C4. Hard ★ Where is the 1s electron?

For hydrogen, $P(r) = \dfrac{4}{a_0^3}\, r^2\, e^{-2r/a_0}$. (a) Show that P is maximum at $r = a_0$. (b) Compute $P(2a_0)/P(a_0)$. (c) Show that the probability of finding the electron inside the sphere of radius a_0 is $1 - 5e^{-2} \approx 32\,\%$ (use $\int x^2 e^{-x}\,\mathrm{d}x = -(x^2 + 2x + 2)\,e^{-x}$). (d) Where is the maximum for He^+?

Answer

(a) $P'(r) = \frac{4}{a_0^3} e^{-2r/a_0}\left(2r - \frac{2r^2}{a_0}\right) = 0$ at $r = a_0$ (positive before, negative after): a maximum.
(b) $\dfrac{P(2a_0)}{P(a_0)} = \dfrac{4a_0^2 e^{-4}}{a_0^2 e^{-2}} = 4e^{-2} = 0.54$.
(c) With $x = 2r/a_0$: $\int_0^{a_0} P\,\mathrm{d}r = \frac{1}{2}\int_0^2 x^2 e^{-x}\,\mathrm{d}x = \frac{1}{2}\left[2 - 10e^{-2}\right] = 1 - 5e^{-2} = 0.323$.
(d) Replace a_0 by a_0/Z: maximum at $a_0/2 = 26.5$ pm.

C5. Hard ★ The 2s orbital and its node

For the hydrogen 2s orbital, $\Psi_{2s} \propto \left(2 - \dfrac{r}{a_0}\right) e^{-r/2a_0}$. (a) Find the radius of the radial node, in pm. (b) The radial density is $P(r) \propto r^2 \left(2 - \frac{r}{a_0}\right)^2 e^{-r/a_0}$: show that its maxima are at $r = (3 \pm \sqrt{5})\,a_0$, and say which one is the most probable distance. (c) Where is the node for Li^{2+}?

Answer

(a) $\Psi_{2s} = 0$ for $r = 2a_0 = 106$ pm (a sphere; $n - \ell - 1 = 1$ node).
(b) Write $P = f^2 e^{-r/a_0}$ with $f = r(2 - r/a_0)$: $P' = e^{-r/a_0} f\,[2f' - f/a_0]$. $f = 0$ gives $r = 0$ and the node $r = 2a_0$ (minima). $2f' = f/a_0$: $4 - 4r/a_0 = 2r/a_0 - r^2/a_0^2$, i.e. $r^2 - 6a_0 r + 4a_0^2 = 0$, $r = (3 \pm \sqrt{5})\,a_0 = 0.76\,a_0$ and $5.24\,a_0$. The outer maximum is the higher one: the most probable distance is $5.24\,a_0 \approx 277$ pm, consistent with $r = n^2 a_0 = 4a_0$ being the size scale.
(c) Replace a_0 by $a_0/3$: $2a_0/3 = 35$ pm.

C6. Moderate Degeneracy and Madelung's order

(a) In hydrogen, how many orbitals have the energy E_3? Give their energy in eV. (b) Why does this degeneracy disappear in a many-electron atom such as Na? Order 3s, 3p, 3d for Na. (c) Use Madelung's rule to order 6s, 4f, 5d, 6p, 5f, 7s, and say which pairs have the same $n + \ell$.

Answer

(a) $n^2 = 9$ orbitals (3s, three 3p, five 3d), all at $E_3 = -13.6/9 = -1.51$ eV.
(b) The other electrons screen the nucleus; s electrons penetrate closer to it than p and d electrons, so they feel a larger charge: $E_{3s} < E_{3p} < E_{3d}$.
(c) $n + \ell$: 6s 6, 4f 7, 5d 7, 6p 7, 7s 7, 5f 8. Equal sums are ordered by n: 6s < 4f < 5d < 6p < 7s < 5f. Same $n + \ell = 7$: 4f, 5d, 6p, 7s.

C7. Moderate Spin and magnetic field

(a) Sodium's D line is a doublet at 589.0 and 589.6 nm. Compute the energy gap between the two upper levels in meV and the wavenumber gap in cm^{-1}. What property of the electron explains it? (b) Into how many sublevels does a 3d level split in a magnetic field (Zeeman effect)? A 4f level? Which quantum number is revealed?

Answer

(a) $\Delta E = hc\left(\frac{1}{589.0\times10^{-9}} - \frac{1}{589.6\times10^{-9}}\right) = 3.43\times10^{-22}$ J $= 2.1$ meV; $\Delta(1/\lambda) = 17.3$ cm^{-1}. The spin ($m_s = \pm\frac{1}{2}$) splits the 3p level into two.
(b) $2\ell+1$ values of m_ℓ: 5 for 3d, 7 for 4f. It reveals the magnetic quantum number m_ℓ.

C8. Moderate Recognise the orbital

Name the orbital: (a) one nodal plane, the (xOy) plane, and no radial node; (b) two angular nodes and one radial node; (c) spherical, with two radial nodes; (d) $\Psi_{3,1,0}$. For (a) and (d), give m_ℓ and the occupied/unoccupied status in ground-state hydrogen.

Answer

(a) $\ell = 1$, $n = 2$, plane (xOy): $2p_z$ ($m_\ell = 0$). (b) $\ell = 2$, $n = 4$: 4d. (c) $\ell = 0$, $n = 3$: 3s. (d) $n = 3$, $\ell = 1$, $m_\ell = 0$: $3p_z$. Both are unoccupied in ground-state hydrogen (only 1s is).

C9. Hard ★ Sizes of hydrogen-like orbitals

Using $r = n^2a_0/Z$: (a) compare the sizes of H 1s, Li^{2+} 2s and Li^{2+} 3d. (b) Which hydrogen-like ion has a 2p orbital the same size as the 1s of hydrogen? Give the energy of that 2p level.

Answer

(a) H 1s: $a_0 = 52.9$ pm. Li^{2+} 2s: $4a_0/3 = 70.5$ pm. Li^{2+} 3d: $9a_0/3 = 3a_0 = 159$ pm.
(b) $n^2/Z = 1$ with $n = 2$: $Z = 4$, Be^{3+}. $E_2 = -13.6\times16/4 = -54.4$ eV.

Part D · Synthesis: chapters 1 to 3 together

D1. Hard ★ Coulomb meets Bohr

In the Bohr picture the electron of hydrogen orbits at $r_n = n^2a_0$. Its total energy is $E = -\dfrac{ke^2}{2r}$ (half its Coulomb potential energy). (a) Compute E for $r = a_0$ in J and eV and compare with E_1. (b) Show that this expression gives $E_n = E_1/n^2$. (c) For He^+ in $n = 2$, give r and E; compare with hydrogen's ground state.

Answer

(a) $E = -\dfrac{8.99\times10^9\times(1.6\times10^{-19})^2}{2\times52.9\times10^{-12}} = -2.18\times10^{-18}$ J $= -13.6$ eV $= E_1$.
(b) $r_n = n^2a_0$ gives $E_n = -\dfrac{ke^2}{2a_0n^2} = \dfrac{E_1}{n^2}$.
(c) Hydrogen-like ion: $r = n^2a_0/Z = 4a_0/2 = 106$ pm; $E = -13.6\times4/4 = -13.6$ eV: the same energy as H in its ground state, but twice as far out.

D2. Hard ★ An unknown ion

A hydrogen-like ion emits a series ending on $n_1 = 2$ whose limit is at 22.8 nm. Identify the ion, then give its ground-state ionisation energy and the size of its 1s orbital.

Answer

Limit: $\nu = Z^2 R_H / n_1^2$, so $Z^2 = \dfrac{c\,n_1^2}{\lambda R_H} = \dfrac{3.00\times10^{8}\times 4}{22.8\times10^{-9}\times 3.29\times10^{15}} = 16.0$: $Z = 4$, the ion is Be^{3+}. $E_I = 13.6\times 16 = 218$ eV; $r_{1s} = a_0/4 = 13.2$ pm.

D3. Hard ★ Photoionisation by a helium lamp

A helium discharge lamp emits UV light at 58.4 nm. (a) Compute the photon energy in eV. (b) It hits ground-state hydrogen atoms: what happens, and what is the kinetic energy and speed of the electron that leaves? (c) Could this lamp ionise He^{+}? Justify.

Answer

(a) $E = hc/\lambda = 3.40\times10^{-18}$ J $= 21.3$ eV.
(b) $21.3 > 13.6$ eV: the atom is ionised. $E_k = 21.3 - 13.6 = 7.7$ eV $= 1.22\times10^{-18}$ J; $v = \sqrt{2E_k/m_e} = 1.64\times10^{6}$ m s^{-1}.
(c) No: He^{+} needs $13.6\times 4 = 54.4$ eV > 21.3 eV.

Unofficial study material